

5.1 Current Business Process

Currently, navigating the huge Universiti Teknologi Malaysia (UTM) campus relies heavily on traditional, manual coordination and fixed timetable shuttle buses. When a student needs to travel from their residential college to a faculty building, they walk to the nearest designated bus stop and wait. Due to there is no live tracking visibility into the transit network, the student must rely on guesswork, often standing under the scorching hot sun or getting soaked during sudden heavy rains, unsure if the bus is two minutes or twenty minutes away.

During peak morning and evening hours, the situation deteriorates. When a bus finally arrives, the student frequently discovers it is already packed with passengers from the previous stops. Unable to board, the student is forced to wait for the next uncertain bus, resulting in a domino effect of missed lectures, late lab sessions, and high “transit anxiety”. If a student decides to seek alternative transport, they must resort to fragmented, unofficial carpooling. They open disorganized social media groups or messaging apps (like Telegram) to manually ask for rides, which is inefficient and poses security concerns as driver identities are not formally verified.

Furthermore, for freshmen or visitors navigating the campus maze, there is no centralized tool to plan their journey. They often take long, inefficient walking paths or miss necessary bus transfers. Finally, when the journey concludes, if the students experience a delayed schedule, reckless driving, or a dirty vehicle, there is no dedicated channel to report it. The administration remains unaware of these daily operational failures, and the cycle of frustration continues into the next semester.

5.2 Functional Requirement

User Requirement Definition

A user requirement definition was conducted to explain the services and operational constraints of the system from the perspective of the UTM students and administration, written in natural language for the stakeholders.

1. The system shall strictly authenticate users using their official UTM email addresses to ensure a secure, student-only environment.
2. The system shall display the exact real-time locations and Estimated Times of Arrival (ETA) for all campus buses on an interactive map.
3. The system shall allow users to monitor the current seat availability of the approaching buses before they arrive at the station.
4. The system shall enable students to securely host or join carpool rides with peers heading in the same direction.
5. The system shall provide an intelligent route planning feature that calculates the fastest path and guides users through campus transit transfers.
6. The system shall allow users to submit transparent feedback, rate their trips, and report specific issues immediately after their journey.

System Requirement Specification

The system requirement specification details the precise functions, services, and operational logic required to build the platform, ensuring clear guidelines for developers and stakeholders.

Live Tracking & Mapping Requirements

- 1.1 The system shall establish a continuous data connection with GPS transponders installed on all campus buses.
- 1.2 The system shall process incoming GPS coordinates and plot the actual location of each bus on the digital campus map in real time.
- 1.3 The system shall calculate precise Estimated Times of Arrival (ETA) based on current bus speed, distance to the user's selected station, and traffic conditions.
- 1.4 The system shall display specific bus details, such as the plate number and assigned route, when a user selects a bus icon on the map.

Seat Availability Processing Requirements

- 2.1 The system shall dynamically update and store the current occupancy level of each active campus bus.
- 2.2 The system shall categorize and visually display the real-time seat status to the user using clear indicators (e.g., Available, Moderate, or Full).
- 2.3 The system shall automatically prompt administrators if a specific route consistently reports a “Full” capacity over a designated time period.

Carpool Management Requirements

- 3.1 The system shall provide separate interfaces for “Drivers” to post available seats and “Passengers” to search for rides.
- 3.2 The system shall match drivers and passengers based on matching departure times and destination proximity.
- 3.3 The system shall facilitate secure communication by displaying verified contact details only after a ride match is confirmed by both parties.
- 3.4 The system shall record a digital log of all carpool transactions, including the identities of the driver and passenger, for campus safety purposes.

Intelligent Route Planning Requirements

- 4.1 The system shall provide an input field for users to enter their current departure point and desired campus destination.
- 4.2 The system shall calculate the most efficient route by factoring in walking distance, active bus route, and potential transfer points.
- 4.3 The system shall generate step-by-step navigation instructions for the user to follow.

Feedback and Reporting Requirements

- 5.1 The system shall automatically trigger a rating prompt (e.g., 1 to 5 stars) on the user’s device once a journey is marked as completed.
- 5.2 The system shall categorize submitted complaints (e.g., “Late Arrival”, “Vehicle Condition”, “Safety Concern”) and store them in the centralized database.
- 5.3 The system shall generate monthly performance reports summarizing student feedback for the university’s transport administration.

5.3 Non-functional Requirement

Non-functional requirements dictate how the system should perform, ensuring a seamless, reliable, and secure experience for thousands of concurrent UTM students. These focus on the platform's quality attributes and operational constraints. These focus on the platform's quality attributes and operational constraints.

Product Requirements

- Performance: The system shall be capable of processing live GPS data and rendering it on the user's map within 3 seconds of user interaction, avoiding any lag that could result in missed buses.
- Reliability: The system shall maintain 99.5% uptime during peak campus operating hours (Monday to Friday, 7:00 AM to 7:00 PM) to ensure students are never stranded without transit data.
- Size and Scalability: The system's cloud backend (AWS/Firebase) shall automatically scale its storage (Mbytes) and processing resources to seamlessly support up to 10000 concurrent users without performance degradation.
- Portability: The platform shall support a wide number of target systems, remaining fully operable on both iOS and Android mobile operating systems.
- Ease of Use: The user interface shall be designed with a "frictionless" approach, minimizing the training time required for a student to navigate the app and view bus ETAs.
- Robustness: The database shall have automated backup protocols. In the event of a system failure, the time to restart after failure shall not exceed 10 minutes, and the probability of data corruption on failure shall be kept below 1%.

Organizational and Process Requirements

- Process Mandates: The system development shall mandate the use of specific Integrated Development Environment (IDE) and cross-platform programming languages (e.g., Flutter) to streamline internal campus IT workflows.
- Administration: The IT team of the university's Transportation Department should be granted a secure, dedicated admin dashboard with its own unique authentication credentials to monitor network health and view feedback reports.
- Training: The system shall operate automatically from the drivers' perspective, requiring zero training time for the campus bus drivers.

External Requirements

- Integration: The system shall reliably integrate with the external Google Maps API to fetch accurate campus topographies and calculate route distances.
- Security and Privacy: The system shall implement strict data encryption (e.g., AES-256) to protect user profiles, real-time location data, and carpool logs, complying with external privacy provisions such as the Personal Data Protection Act (PDPA).
- Authentication: Users of the system shall authenticate themselves by interfacing with UTM's official domain registry to ensure that only verified emails ending in @graduate.utm.my or @utm.my can successfully register.